KAND CA at NADI 2026 Shared Task: Multi-Stage Fine-Tuning for Spoken Arabic Dialects

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Abstract

We describe the KAND CA systems for all five subtasks of the NADI 2026 shared task on multidialectal Arabic speech processing. For the three ASR subtasks, a single family of *staged, fully fine-tuned* Whisper large-v3 models — broad dialectal adaptation, then only task-matched gold data — is combined with task-specific levers: an mT5 spelling-correction post-editor trained on the task references (Subtask 1.1, final country-average WER **0.5111**), chained continue-fine-tuning through the Subtask 1.1 data before the Mixed-ASR dev split (Subtask 1.2, WER **0.48**), and a fresh two-epoch schedule over a 46K-clip external gold Tunisian–French corpus with an accent-only French post-pass (Subtask 1.3, WER **15.21%**). For spoken dialect identification (Task 2), where in-domain accuracy *anti*-predicted blind-test accuracy, our system fuses an architecturally distinct FastConformer-encoder classifier, a decorrelated enhanced-audio view, and an ASR-transcript text route, reaching **0.5114** accuracy. For dialectal TTS (Subtask 3) we submitted 4,001 voice-cloned clips across ten dialects (UTMOS 2.72). We also report extensive negative results and release our models, correctors, and code publicly.

1 Introduction

Arabic speech processing is complicated by diglossia and by wide phonological, lexical, and orthographic variation across national dialects: systems trained predominantly on Modern Standard Arabic (MSA) degrade sharply on dialectal speech. The NADI 2026 shared task (Sullivan et al., 2026), the second multidialectal Arabic *speech* edition of the NADI series (Talafha et al., 2025), targets this failure mode across five subtasks: robust multi-dialect ASR (1.1), mixed-dialect ASR (1.2), Tunisian code-switched ASR (1.3), 20-way spoken dialect identification (Task 2), and dialectal TTS (Subtask 3).

Our strategy has two pillars. For ASR, we *stage* the adaptation of Whisper large-v3 (Radford et al., 2023): a broad dialectal-mixture full fine-tune first, then successive gentle continue-fine-tunes on *task-matched gold data only*, with task-specific final levers (learned spelling correction, chained fine-tuning, disciplined scheduling). For dialect identification, where the blind test is severely out-of-domain, we accumulate decorrelated *information routes* at conservative weights — an architecturally alien encoder, an enhanced-audio view, then a text route — rather than maximizing in-domain fit.

Our final systems scored 0.5111 country-average WER (1.1), 0.48 WER (1.2), 15.21% WER (1.3), and 0.5114 accuracy (Task 2). Beyond the recipes, we contribute a map of what does *not* work: dev-set signals that invert on the blind test, silver labels, specialist routing, seven collapsing dialect-ID backbones, test-side enhancement, decoding search, and a fluent LLM post-editor costing 5.9 WER points. Our models, correctors, probe sets, and training/inference scripts are public.[1]

2 Background

Table 1 summarizes the five subtasks (we follow the official subtask numbering). Subtask 1.1 covers Algeria, Egypt, Jordan, Mauritania, Morocco, Palestine, UAE, and Yemen; submissions are one text file per dialect in dataset row order. Subtask 1.2 has *no train split* — its 3,152-clip dev set is, to our knowledge, the only labeled mixed-dialect Arabic ASR data available anywhere; clips run up to ∼4 minutes. Subtask 1.3 uses the TEDx-domain `NADI_TUN_ASR_2026` corpus (Bougares, 2026), in which 73% of transcripts code-switch into French or English; its scorer keeps Latin accents (French *château* ≠ *chateau*) and does not normalize Arabic orthographic variants, so the gold *convention* is scored, not just content. Task 2 uses the ADI-20 label set (MSA plus 19 country dialects), extending

[1] https://huggingface.co/oddadmix; TTS code: https://github.com/Oddadmix/Lahgtna-OmniVoice.

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Task	Input → Output	Official data	Metric	Blind test
1.1 Robust ASR	8-dialect speech → text	23.6K train / 10.8K val	country-avg WER, CER	8×500 clips
1.2 Mixed ASR	mixed-dialect speech → text	3,152 dev (no train)	WER	427 clips
1.3 Code-switched ASR	Tunisian+FR/EN → text	24.9K train / 731 val	WER	2,123 clips
Task 2 Dialect ID	speech → 1-of-20 label	ADI-20	accuracy, C_{avg}	878 clips
Subtask 3 TTS	text + dialect → speech	— (test prompts only)	UTMOS, NISQA, WER	4,001 prompts

Table 1: The five NADI 2026 subtasks we participated in. See Sullivan et al. (2026) for full task details.

ADI-17 (Shon et al., 2020), and scores accuracy plus the NIST LRE detection cost over submitted logits.

Input/output examples. ASR maps a 16 kHz clip to an Arabic-script transcript (romanized here); an illustrative 1.3-style reference mixes a Tunisian matrix with French islands — *"bech naàmlou* le suivi *fi el-*projet*"* — the gold attaching the article *el-* to Latin tokens. Task 2: clip → one of 20 labels (e.g. `Morocco`) + logits; Subtask 3: (dialect, text) → 24 kHz waveform.

Whisper-scale models provide strong Arabic substrates but skew toward MSA (Radford et al., 2023); Casablanca (Talafha et al., 2024) showed the value of gold dialectal supervision for exactly the 1.1 dialect set.

3 System Overview

3.1 Shared ASR backbone

All three ASR systems start from `whisper-large-v3-arabic-dialectal-v4`, our full fine-tune of Whisper large-v3 (1.5B) on a broad dialect-balanced mixture: our public lahgtna corpus (KAND CA, 2026a) (54.6K YouTube-sourced clips / 267 h, balanced over 13 dialects), Common Voice 18 Arabic (Ardila et al., 2020), FLEURS (Conneau et al., 2023), and MASC (Al-Fetyani et al., 2021). This shifts the model's prior away from MSA before any task data is seen (contamination audit: Appendix B). All fine-tunes are *full* fine-tunes (cross-entropy, bf16, gradient checkpointing); continue-fine-tune rounds use lr 5×10^{-6} with a paged 8-bit AdamW (Dettmers et al., 2022) and best-on-holdout checkpoint selection. Long-form decoding uses a chunked pipeline (30 s chunks, 5 s stride) with a post-hoc *repeat-collapse* pass that caps immediately repeated 1–4-grams at two copies; this recipe is essential (naive batched generation inflates 1.1 dev WER 0.579 → ≈0.80).

3.2 Subtask 1.1: staged fine-tuning + learned spelling correction

The acoustic model is staged through: (i) the dialectal base above; (ii) full fine-tune on the concatenated task material (`nadi2026-full-v4`; blind 0.57); (iii) continue-fine-tunes folding in Casablanca gold (its released val+test data — it ships no train split — as permitted external resources; blind 0.56), targeted Algerian conversational data (kahwa-postcast, ∼7.2K French code-switched clips; blind 0.5576), the official train set with Casablanca replay (blind 0.52), and finally the *full* 13,637-clip Casablanca corpus (in-domain holdout 0.4144 → 0.4021; blind 0.52).

The final gain comes in *text space*: mT5-large (Xue et al., 2021) spelling correctors fine-tuned on synthetic corruptions of *the task references themselves*, applied to every hypothesis. Eight per-dialect correctors yield **0.5111** (from 0.52); a single multidialect corrector ties within leaderboard resolution (reported 0.51) while being 8× simpler. Trained on the task references, the correctors learn the gold orthographic convention (hamza seats, alif forms, ta-marbuta) — the property that makes or breaks ASR post-editing (§B).

3.3 Subtask 1.2: chained continue-fine-tuning

The winning system *chains* adaptation: the model must first be good at 1.1, and only then see the Mixed-ASR data. With no train split shipped, we adapt on the dev split itself (the released labeled data): continue-fine-tuning our best 1.1 model on it (3,002 train / 150 holdout) plus replay (1,500 official-1.1 + 1,000 Casablanca clips) against forgetting. This scores **0.48** (base zero-shot: 0.51); notably, the same 1.1-helpful data applied *directly* to 1.2 regressed to 0.52. Checkpoint selection transferred: the best-on-holdout checkpoint beat the final one on the blind test (0.48 vs. 0.49). One operational hazard dwarfs modeling: the headerless prediction file's line order is *split-specific* (dev sorted-by-ID; test in parquet order); the wrong order scores ≈1.4.

3.4 Subtask 1.3: external gold + disciplined scheduling

From the dialectal base (zero-shot 87.6% WER on the public test split), in-task full fine-tuning with SpecAugment (Park et al., 2019) ($p_{\text{time}} = p_{\text{feat}} = 0.05$) reaches 20.5. The decisive ingredient is `tunisian-asr-cleaned` (FARUKx-AUTO, 2026), 46K external gold Tunisian–French code-switched clips: mixed with a TEDx replay (70,595 clips total) it yields 15.31 after one epoch. Scheduling then matters more than data: warm-continuing one more epoch *regressed* (15.4), whereas re-running *fresh* with a single 2-epoch cosine schedule won (`faruk-v7`, **15.22**); four epochs overfit. A final post-pass restores French accents via a spellchecker with an accent-only filter (`unaccent(fix)==token`, provably unable to change letters); it touched 5 tokens in 2,123 clips for the final **15.21**.

3.5 Task 2: decorrelated routes for OOD dialect ID

Our development overturned its own assumptions at each phase. *Phase 1*: a four-way softmax ensemble of MMS (Pratap et al., 2024), w2v-BERT 2.0 (Chung et al., 2021; Seamless Communication team, 2023), Whisper-encoder, and ECAPA-TDNN (Desplanques et al., 2020; Valk and Alumäe, 2021) classifiers reached 95.21% on a 10,806-clip holdout — and collapsed to 0.22–0.33 on the blind test, in *reverse* order of holdout strength. *Phase 2*: a single attention-pooled classifier on the FastConformer (Rekesh et al., 2023) encoder of `cohere-transcribe-arabic` (Cohere Labs, 2026) — weaker in-domain (80.07%) — scored **0.48**; fusing its raw-audio posteriors with those on audio processed by LavaSR (YatharthS, 2025), a reconstruction-style enhancement model — a decorrelated test-time view (61% agreement) — reached **0.50**. Seven alternative backbones all collapsed (0.05–0.33), and a higher-capacity head on the winning encoder raised in-domain accuracy to 84.26% while *cutting* blind accuracy to 0.38 — the in-domain/OOD inversion reproduced within a single encoder. *Phase 3*: we add a **text route** — transcribe each test clip with our 1.1 ASR system, classify the transcript with a BERT (Devlin et al., 2019) dialect classifier trained on raw dialectal text (15 labels, of which 14 map into ADI-20; unmapped classes receive no text mass), and fuse:

$$\mathbf{p} \propto \tfrac{1}{2}\,\mathbf{p}_{\text{raw}} + \tfrac{1}{2}\,\mathbf{p}_{\text{lava}} + w\,\mathbf{p}_{\text{text}}, \qquad (1)$$

with $\mathbf{p}_{\text{raw}}, \mathbf{p}_{\text{lava}}$ the classifier's posteriors on raw/enhanced audio and $\mathbf{p}_{\text{text}}$ the mapped text posterior. Though the text argmax agrees with the audio on only 16.7% of covered clips, w=0.30 reaches **0.5114**, with a sharply peaked response (§5). Counterintuitively, feeding the text route *better* (corrector-post-edited) transcripts scored 0.509: the text classifier was trained on raw dialectal text, and distribution match beats transcript quality.

3.6 Subtask 3: dialectal voice-cloning TTS

We submitted 4,001 clips across the ten test dialects from `lahgtna-omnivoice-v2` (KAND CA, 2026b), our open 612M-parameter voice-cloning dialectal Arabic TTS model (24 kHz). One reference speaker per dialect drives all clips of that dialect; six of the thirteen dialect references are human-curated (verified audio with accurate transcripts), the rest use demo-space references with automatic transcription. Two unsupported test dialects fall back to their nearest supported neighbor (UAE→Saudi, Jordan→Palestinian). Reference quality matters: a faulty auto-transcribed reference inflated one dialect's outputs to 12.8 s per clip, vs. 2.2 s with a corrected reference. Official automatic evaluation: aggregate UTMOS 2.72 (Saeki et al., 2022), NISQA-MOS 3.57 (Mittag et al., 2021), WER 0.32 / CER 0.11 via Omnilingual ASR 7B (Meta AI, 2025); per-dialect: Table 6.

4 Experimental Setup

All ASR/DID models were trained on a single 32 GB GPU with PyTorch and Transformers (Wolf et al., 2020) (versions and URLs in Appendix A, Table 3); full-parameter large-v3 requires the paged 8-bit optimizer to fit. In-task fine-tunes use lr 1×10^{-5}; continue-fine-tunes lr 5×10^{-6} with `load_best_model_at_end` on an in-domain holdout. mT5 correctors are trained in fp32 (mT5 is numerically unstable in bf16) with an instruction prefix, decoding greedy/beam-4. Dialect-ID classifiers use two-phase training (frozen-encoder head warm-up, then full fine-tune) with class-weighted cross-entropy on the 67K-clip curated ADI-20 subset. We used official scorers and split conventions throughout (jiwer served only for local proxy evaluation); all blind-test numbers below are official leaderboard evaluations, with post-submission results marked. Each progression row in Table 2 was one official blind submission, plus a few additional variants per task. Full hyperparameters are

Subtask 1.1 system	WER
Stage-2 task FT (`nadi2026-full-v4`)	0.57
+ Casablanca continue-FT	0.56
+ Algerian conversational data	0.5576
+ official train data (+ replay)	0.52
+ full Casablanca (submitted acoustic)	0.52
+ 8 per-dialect mT5 correctors	**0.5111**
+ single multidialect corrector	0.51

Subtask 1.2 system	WER
base, zero-shot + decode recipe	0.51
1.1-helpful continue-FTs, direct	0.52
chained Mixed-ASR FT (best ckpt)	**0.48**
same, final checkpoint	0.49
+ SADA gold / + correctors	0.48

Subtask 1.3 system	WER%
in-task full FT + SpecAugment	20.5
+ external gold, 1 ep warm	15.31
fresh 2-epoch schedule (`faruk-v7`)	15.22
+ French accent restoration	**15.21**
+ Qwen LoRA corrector (post-sub.)	21.15

Task 2 system	Acc
4-way in-domain ensemble (95.21% holdout)	0.33
FastConformer classifier, solo	0.48
+ LavaSR test-time view	0.50
+ text route, w=0.15	0.5057
+ text route, w=0.30 (submitted)	**0.5114**
+ text route, w=0.50	0.50

Table 2: Official blind-test progression for all four scored subtasks. All rows are official evaluations except where marked post-submission. WER is shown as a fraction for 1.1/1.2 and as percent for 1.3, following each leaderboard's convention; accuracy as a fraction.

in Appendix A.

5 Results

Table 2 gives the blind-test progression of every submitted system. Caveats: the blind sets are small (427–4,000 clips), so third-decimal differences span a handful of clips — we treat deltas $\leq$0.01 as ties; the Task 2 leaderboard ranked accuracy and did not expose per-submission C_{avg} (in-domain ensemble: 0.0170).

Only task-matched gold transfers. Across 1.1 and 1.2, every blind-test win came from official task data or organizer-released gold (Casablanca); silver labels, off-distribution "new dialect data," and adjacent-dialect gold (SADA Khaliji, SDAIA, 2022; holdout win, blind tie) improved proxies but not the test. The dev signal *inverted* three times on 1.1: both decisive interventions (Casablanca, the correctors) looked neutral-to-worse locally yet won on the blind test.

Error analysis: convention, not content. The dominant recoverable error class in dialectal Arabic ASR is orthographic convention. On 1.1, the correctors' wins come from normalizing hamza seats, alif variants, and ta-marbuta toward the reference convention (809/4,000 hypotheses changed); on 1.3, the same lever *reverses*: a Qwen3.5-4B LoRA corrector (Qwen Team, 2026b; Hu et al., 2022) whose edits looked excellent on inspection — completing truncated French (*communaut* $\rightarrow$ *communauté*), fixing agreement (*ignorer* $\rightarrow$ *ignore*) — cost 5.9 WER points (15.21 $\rightarrow$ 21.15), because a model fine-tuned on the task gold already emits the reference convention, so a differently-conventioned corrector can only move hypotheses away from it. The second error class is long-form repetition loops: post-hoc repeat-collapse recovers $\approx$2.7 WER points on 1.2, while n-gram blocking *hurts* (it deletes legitimate Arabic reduplication). On Task 2 the audio and text routes agree on only 16.7% of covered clips yet fusion gains a point — their errors are largely disjoint; decorrelation, not accuracy, pays.

In-domain accuracy anti-predicts OOD. On Task 2 the holdout ranking of systems is the reverse of the blind-test ranking (95.21% $\rightarrow$ 0.33 vs. 80.07% $\rightarrow$ 0.48), across encoders, data scales (a 160K cross-source model: holdout 94.54%, blind 0.22), and head capacities. Audio "cleaning" consistently hurts dialect ID — channel and prosodic texture carry dialect signal — though the same enhancer helps as a *decorrelated second view*.

6 Conclusion

We presented a family of staged, fully fine-tuned Whisper systems for three Arabic ASR subtasks (0.5111, 0.48, 15.21% WER), a decorrelated multi-route fusion for out-of-domain dialect identification (0.5114 accuracy), and a voice-cloned dialectal TTS submission (UTMOS 2.72). The lessons: feed staged adaptation only task-matched gold; treat output order and orthographic convention as first-class components; distrust local proxies; under domain shift, add routes rather than fit. The open problems are data problems — Mixed ASR is bound by its single 3,152-clip labeled set; Mauritanian ASR stayed near 0.78 WER throughout — and call for collection, not adaptation.

Limitations

Our systems are tuned to the NADI 2026 orthographic and scoring conventions; the correctors in particular are convention-bound by design and will not transfer to differently-transcribed references. Several final choices (corrector on/off, fusion weight w) were selected on the leaderboard because local dev signals were demonstrably unreliable — at the margin the blind test served as our development set (submission counts: Table 2). Compute was one 32 GB GPU, so ablations were run once rather than over multiple seeds. The YouTube-sourced lahgtna corpus is used under research terms. The TTS fallbacks for unsupported dialects cannot capture those dialects' actual phonology.

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A Hyperparameters and Tooling

Tooling: PyTorch 2.11 (`https://pytorch.org`), Transformers 5.13 (`https://github.com/huggingface/transformers`), bitsandbytes 0.49 (`https://github.com/bitsandbytes-foundation/bitsandbytes`), jiwer 4.0 (`https://github.com/jitsi/jiwer`), LavaSR (`https://huggingface.co/YatharthS/LavaSR`). The silver-label experiments and the 1.2 label audit used Gemini 3.5 Flash (Google DeepMind, 2026).

B Negative Results

Table 4 catalogs the interventions that tied or regressed. Systems referenced there: CrisperWhisper (Zusag et al., 2024), AraT5 (Nagoudi et al., 2022), Qwen3-ASR (Qwen Team, 2026a), and Gemini 3.5

Setting	Value
ASR base	Whisper large-v3 (1.5B)
Precision	bf16 + grad. checkpointing
In-task FT lr	1×10^{-5}
Continue-FT lr	5×10^{-6}, cosine
Optimizer	paged AdamW 8-bit
Batch	$4 \times$ accum. 4 (eff. 16–32)
SpecAugment (1.3)	$p_{\text{time}}{=}p_{\text{feat}}{=}0.05$
Checkpoint rule	best on in-domain holdout
Decoding	greedy; chunked 30 s/5 s stride
De-looping	collapse 1–4-gram repeats >2
Correctors	mT5-large, fp32, beam 4
DID training	frozen-head then full; class-weighted CE
DID fusion	$0.5/0.5$ raw/LavaSR $+ w \cdot$ text
Software	PyTorch 2.11, Transformers 5.13, bitsandbytes 0.49, jiwer 4.0
Hardware	$1\times$ 32 GB GPU

Table 3: Shared hyperparameters across systems.

Flash (Google DeepMind, 2026). Two audits deserve note: (i) an audio-hash audit (MD5 over raw bytes and decoded PCM) found *zero* overlap between the 1.3 blind set and the public Tunisian corpora, so training on public test splits was never going to leak signal; (ii) a full Gemini re-transcription of the 1.2 dev set found zero clips whose gold is incoherent while Gemini's transcript is coherent (median CER of Gemini hypotheses against the gold references: 0.150), ruling out label noise as the cause of the 0.48 plateau.

C Data Inventory

Table 5 lists the principal training resources with sizes; all are public.

D Per-Dialect Results

Subtask 1.1. The per-dialect spread is large and stable across systems: Palestine, Jordan, and UAE are easiest (≈0.41–0.44 WER for the Stage-2 model); Yemen, Egypt, and Morocco intermediate; and the Maghrebi/Saharan tail hardest — Algeria improved $0.71 \rightarrow 0.662$ with targeted conversational data, while Mauritania stayed at 0.78–0.79 under every intervention we tried, making it the dominant open problem for this dialect set.

Subtask 3. Table 6 gives the official per-dialect automatic evaluation. Quality prediction and intelligibility diverge: Morocco and Sudan score highest on UTMOS yet Morocco has the worst WER, while Saudi and Palestine pair mid UTMOS with the best WER; Yemen is weakest on every metric, consistent with its reference being auto-transcribed.

Intervention	Outcome (blind)
Subtask 1.1	
Gemini silver labels	regressed dev + blind
New Algerian YT data	dev win, blind 0.52→0.53
Per-dialect routing (×3)	0.52 vs. 0.5111
Cohere ASR (LoRA FT)	val 0.665 vs. ours 0.325
CrisperWhisper base	holdout 0.519 vs. 0.402
Test-side enhancement	0.52→0.55–0.88
Subtask 1.2	
SADA-Khaliji gold	holdout win, blind tie
mT5 / Qwen correctors	tie (0.48)
Qwen3-ASR native FT	0.59
Gemini label audit	0 broken-gold clips
Subtask 1.3	
3rd/4th epoch; warm cont.	15.4 / overfit
Egyptian–English CS data	15.42
"Cleaner" re-processing	val 19.44 vs. 17.26
Beam/n-gram-block decode	tie to +4pt val
AraT5 / mT5 / Qwen corr.	+1.4 / tie / +5.9
Public-test-split training	no transfer (0 hash overlap)
Task 2	
7 alt. backbones	0.05–0.33
Higher-capacity head	0.48→0.38
160K cross-source data	0.22
Extra TTA/enhanced views	dilute to 0.44–0.49
Corrected transcripts	0.509 vs. 0.5114

Table 4: Negative results across subtasks; each row ties or regresses relative to the submitted system.

Resource	Clips
NADI 1.1 official train	23,608
Casablanca (gold, 8 dialects)	13,637
Mixed-ASR dev (only labeled 1.2 data)	3,152
NADI_TUN train	∼24.9K
tunisian-asr-cleaned (external gold)	∼46K
ADI-20 curated subset	67K
lahgtna (base mix, 267 h)	54,600
CV18 + FLEURS + MASC (base mix)	—
SADA22-Khaliji (probe; no blind gain)	6K

Table 5: Principal training resources (KAND CA, 2026a; Talafha et al., 2024; Bougares, 2026; FARUKx-AUTO, 2026; Shon et al., 2020; Ardila et al., 2020; Conneau et al., 2023; Al-Fetyani et al., 2021; SDAIA, 2022).

Dial.	N	UTMOS	N-MOS	N-TTS	WER	CER
ae	537	2.73	3.71	3.45	0.313	0.155
dz	293	2.60	3.81	2.96	0.446	0.130
eg	424	2.81	2.75	3.75	0.205	0.054
jo	631	2.60	4.21	3.18	0.328	0.082
ma	201	3.23	3.52	3.05	0.555	0.216
ps	750	2.58	4.23	3.09	0.202	0.057
sa	100	2.79	3.85	3.45	0.143	0.049
sd	114	3.19	4.46	3.98	0.357	0.100
tn	623	3.04	2.74	3.20	0.445	0.122
ye	328	2.17	2.65	2.46	0.479	0.251
all	4,001	2.72	3.57	3.21	0.325	0.114

Table 6: Official Subtask 3 automatic evaluation per dialect: UTMOS (Saeki et al., 2022), NISQA-MOS and NISQA-TTS (Mittag et al., 2021), and WER/CER computed by the organizers with Omnilingual ASR 7B (Meta AI, 2025).